# Technical Documentation

This report documents the workings of the scripts used to complete the assessment. The project was performed using the OpenCV library and the Python programming language. The objective is to complete two tasks: basic image manipulation and creating a color detector using a webcam.

## Task 1: Basic Image Manipulation

The script performs a series of transformations on an input image, including grayscale conversion, Gaussian blurring, and Canny edge detection. The processed images are saved to an output directory and displayed for visualization. This task aims to load any image and perform grayscale conversion, Gaussian blurring, and Canny edge detection on that image. The output images are saved in a separate folder. The script follows a sequential image processing workflow.

- Image reading: The **readImage()** functions load the image. Note that the image should be in the same folder as the Python file. The program will not execute if the image is not found.
- The **convert_to_grayscale()** function converts the BGR (Blue-Green-Red) image to grayscale using **cv.COLOR_BGR2GRAY**. The grayscale image is saved as **grey_img.jpg** and displayed in a window.
- The **apply_gaussian_blur()** function applies a Gaussian filter to smooth the image and reduce noise. The default kernel size is (7, 7) with a sigma (σ) of 0. The blurred image is saved as **gaussianblur.jpg** and displayed.
- The **detect_edges()** function applies the Canny edge detector with default thresholds (50 and 100), and the output is saved and displayed.

The task is not perfect and has room for improvement. The first major improvement would be to handle the input dynamically and enhance error handling. The second would be to extend the script to handle multiple images.

## Task 2: Color Detector

The application captures live video from a webcam and allows users to click anywhere on the video feed to instantly retrieve the precise BGR color values at the selected pixel location. The script provides immediate BGR color feedback from live video and displays color information both visually and in the console. The core components of the script are explained as follows:

- The **VideoCapture** module handles webcam initialization and frame capture.
- The **Mouse Event Handler(setMouseCallback)** Registers left-click events on the display window and captures pixel coordinates, and retrieves BGR values
- The **Information Display System** implements a semi-transparent overlay for clean text display and shows both position coordinates and BGR values.

The main challenge that was faced during this project was to display the closest color name. The immediate improvements would be to implement multiple color space displays using a library like **webcolors**.